

Arithmetic Formulas

Profit and Loss – Formulas

IMPORTANT FACTS

Cost Price:

The price, at which an article is purchased, is called its cost price, abbreviated as C.P.

Selling Price:

The price, at which an article is sold, is called its selling prices, abbreviated as S.P.

Profit or Gain:

If S.P. is greater than C.P., the seller is said to have a profit or gain.

Loss:

If S.P. is less than C.P., the seller is said to have incurred a loss.

IMPORTANT FORMULAE

1. $\text{Gain} = (\text{S.P.}) - (\text{C.P.})$

2. $\text{Loss} = (\text{C.P.}) - (\text{S.P.})$

3. Loss or gain is always reckoned on C.P.

4. Gain Percentage: (Gain %)

$$\text{Gain \%} = \left(\frac{\text{Gain} \times 100}{\text{C.P.}} \right)$$

5. Loss Percentage: (Loss %)

$$\text{Loss \%} = \left(\frac{\text{Loss} \times 100}{\text{C.P.}} \right)$$

6. Selling Price: (S.P.)

$$SP = \left[\frac{(100 + \text{Gain \%})}{100} \times C.P. \right]$$

7. Selling Price: (S.P.)

$$SP = \left[\frac{(100 - \text{Loss \%})}{100} \times C.P. \right]$$

8. Cost Price: (C.P.)

$$C.P. = \left[\frac{100}{(100 + \text{Gain \%})} \times S.P. \right]$$

9. Cost Price: (C.P.)

$$C.P. = \left[\frac{100}{(100 - \text{Loss \%})} \times S.P. \right]$$

10. If an article is sold at a gain of say 35%, then S.P. = 135% of C.P.

11. If an article is sold at a loss of say, 35% then S.P. = 65% of C.P.

12. When a person sells two similar items, one at a gain of say $x\%$, and the other at a loss of $x\%$, then the seller always incurs a loss given by:

$$\text{Loss \%} = \left(\frac{\text{Common Loss and Gain \%}}{10} \right)^2 = \left(\frac{x}{10} \right)^2.$$

13. If a trader professes to sell his goods at cost price, but uses false weights, then

$$\text{Gain \%} = \left[\frac{\text{Error}}{(\text{True Value}) - (\text{Error})} \times 100 \right] \%.$$